

Batch Production Record: Selective Laser Melting of Ti6Al4V Femoral Stems - Batch 20

Batch ID: BPR-SLM-2024-001

Product: Ti6Al4V Femoral Stems (Medical Implant Grade)

Quantity: 20 units

Build Date: June 10-12, 2024

Operator: Maria Rodriguez (Certification ID: OP-SLM-042)

Equipment: SLM Solutions 280HL, Serial No. SLM-2023-045

Build Platform: Platform #PLT-24B, Build Volume: 280 x 280 x 365 mm

Quality Standard: Internal Medical Device Protocol IMP-07 Rev.3

1.0 Batch Summary and Key Parameters

This document records the complete production cycle for batch BPR-SLM-2024-001, comprising 20 medical-grade titanium femoral stems manufactured using Selective Laser Melting technology. All operations were conducted under controlled atmosphere conditions in Class 8 cleanroom environment.

Build Configuration:

- Part orientation: 45° from vertical axis
- Support structure type: Tree supports with breakaway features
- Layer thickness: 30 microns
- Hatch spacing: 120 microns
- Scan speed: 800 mm/s

Critical Process Parameters:

- Laser power setting: 400 W continuous operation
- Machine average power draw during build: 2.4 kW
- Total build duration: 61.35 hours
- Build chamber oxygen level maintained: < 500 ppm

2.0 Chronological Production Log

2.1 Pre-Build Preparation

2024-06-10 07:30 - Operator M. Rodriguez initiated pre-build checklist

- Machine status: PASS - All systems nominal
- Build platform inspection: PASS - Within flatness tolerance (≤ 0.1 mm)
- Recirculating filter system: PASS - Filter pressure drop within limits
- Gas supply verification: PASS - Argon purity 99.999% confirmed

- 2024-06-10 08:15** - Powder loading procedure
- Material: Ti6Al4V ELI Grade 23, Gas Atomized
 - Powder lot number: TIPOW-240501-A
 - Powder sieving: Through 63µm mesh, moisture content < 0.02%
 - Powder loaded to feed modules: 20.83 kg recorded

- 2024-06-10 09:00** - Build file upload and verification
- Build job: FEM_STEM_20_V3.slj
 - Layer count: 2,045 layers estimated
 - Support structure volume: Approximately 8% of total build volume
 - Build preview simulation: No collisions detected

2.2 Build Execution Phase

- 2024-06-10 10:00** - Build initiation sequence
- Chamber evacuation: To 0.1 mbar completed
 - Argon chamber flooding: 3.03 kg consumed to achieve 900 mbar chamber pressure
 - Oxygen level confirmation: 387 ppm - Within specification
 - Platform heating: To 200°C established

- 2024-06-10 10:30** - Layer deposition commenced
- First layer calibration: PASS - Recoater blade clearance verified
 - Laser calibration: PASS - Beam profile within specification

Build Progress Monitoring Log:

Timestamp	Layer Number	Chamber O ₂ (ppm)	Notes
2024-06-10 12:00	150	412	Normal operation
2024-06-10 18:00	450	398	Intermediate quality scan - PASS
2024-06-11 06:00	850	421	Shift change - Operator J. Chen
2024-06-11 18:00	1,250	435	Powder replenishment - Feed module 2
2024-06-12 06:00	1,650	408	Final quality scan initiated
2024-06-12 23:21	2,045	395	Build completion - Layer 2,045

- 2024-06-12 23:21** - Build termination
- Total build duration: 61.35 hours elapsed
 - Argon consumption during build phase: 25.94 kg maintained for atmosphere control
 - Machine status at completion: All systems nominal

2.3 Post-Process Operations

- 2024-06-13 07:30** - Build chamber cooling and depowdering
- Chamber cooldown: To 30°C achieved over 8 hours
 - Loose powder recovery: 18.99 kg collected for recycling
 - Powder sieving: Through 63µm mesh - Particle size distribution within spec
 - Recovered powder lot assignment: TIPOW-REC-240613-A
- 2024-06-13 10:15** - Part removal and support separation
- Build platform removal: Platform #PLT-24B transferred to post-processing station
 - Support structure removal: Using manual breaking tools
 - Support structure mass: 0.019 kg collected for titanium recycling
 - Visual inspection: All 20 stems intact, no visible cracks or deformations
- 2024-06-13 11:30** - Filter system maintenance
- Process filter replacement: Standard procedure after build completion
 - Filter-captured powder: 0.0208 kg collected - Sent to approved waste management
 - Filter lot: FLT-2024-Q2-18 - Disposed per hazardous material protocol
- 2024-06-13 12:00** - Final parts cleaning and initial measurement
- Ultrasonic cleaning: Isopropyl alcohol bath, 15 minutes
 - Dimensional spot check: 3 samples within ±0.1 mm tolerance
 - Mass verification: Sample stems approximately 88.5 grams each

3.0 Material and Resource Consumption

3.1 Raw Material Inputs

Material Type	Quantity	Lot Number	Purpose
Ti6Al4V Powder (GA)	20.83 kg	TIPOW-240501-A	Primary build material
Argon Gas (Chamber Flood)	3.03 kg	ARG-HP-240609	Initial atmosphere establishment
Argon Gas (Build Phase)	25.94 kg	ARG-HP-240609	Process atmosphere maintenance

Reference: Last batch (BPR-SLM-2024-000) powder usage: 20.75 kg for 20 stems

3.2 Energy Consumption Parameters

Parameter	Value	Unit	Measurement Method
Machine Average Power	2.4	kW	Power analyzer PWR-MTR-04
Total Build Time	61.35	hours	Machine internal timer
Laser Operating Power	400	W	Laser controller reading
Auxiliary Systems Power	1.2	kW	Estimated average

Note: Total electrical energy consumption can be calculated from power and time parameters. Industry benchmark for similar builds: ~2.8 kW average power.

3.3 Output Summary

Output Category	Quantity	Mass (kg)	Disposition
Finished Femoral Stems	20 units	1.77	To heat treatment batch HT-240614
Recoverable Powder	1 lot	18.99	Recycled - Lot TIPOW-REC-240613-A
Support Structures	1 batch	0.019	Titanium recycling stream
Filter Waste	1 filter	0.0208	Landfill disposal per protocol

4.0 Quality Control Checkpoints

Visual Inspection Results:

- All 20 stems: PASS - No visible surface defects
- Support attachment points: ACCEPTABLE - Minimal surface marking
- Powder adherence: WITHIN LIMITS - Standard cleaning required

Dimensional Sampling (3 parts):

- Stem length: 152.3 mm ±0.08 mm (Spec: 152.0-152.5 mm)
- Neck diameter: 14.1 mm ±0.05 mm (Spec: 14.0-14.2 mm)
- Mass consistency: 88.3-88.7 grams per stem

Build Integrity Verification:

- Layer adhesion: CONFIRMED - No delamination observed
- Support removal: SUCCESSFUL - No part damage
- Powder recovery rate: 91.2% of input powder recovered

5.0 Equipment Performance and Maintenance Log

SLM Machine SLM-2023-045 Performance:

- Laser uptime: 99.8% of build duration
- Recoater system: Zero faults recorded
- Temperature stability: $\pm 3^{\circ}\text{C}$ throughout build
- Gas flow consistency: Within 5% of setpoint

Maintenance Performed Post-Build:

- Filter replacement: COMPLETED
 - Optics inspection: SCHEDULED for next maintenance cycle
 - Recoater blade: Within wear limits - Next inspection at 500 build hours
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6.0 Batch Yield and Efficiency Metrics

Material Utilization:

- Powder conversion to product: 8.5% of input mass
- Powder recycling efficiency: 91.2% recovery rate
- Support structure ratio: 1.1% of built mass

Production Rate:

- Build time per stem: 3.07 hours average
- Machine utilization: 85.4% of available build volume
- Total active build time: 61.35 hours

Comparative data: Previous batch (BPR-SLM-2024-000) build time: 62.1 hours for 20 stems

7.0 Approvals and Release

Operator Verification:

Maria Rodriguez, 2024-06-13 14:00

- All process steps completed per procedure
- Documentation verified complete
- Materials accountability confirmed

Quality Assurance Review:

James Chen, QA Inspector ID: QA-028, 2024-06-13 15:30

- Batch record review: ACCEPTED
- Quality checkpoints: ALL PASSED
- Release authorization: GRANTED

Next Process Step:

Parts transferred to Heat Treatment Batch HT-240614 for stress relief and aging cycle.

END OF BATCH PRODUCTION RECORD

Record Closed: 2024-06-13 16:00

Archival Location: Digital Repository SLM-BPR-2024 / Physical Copy Bin 24-06